

Title: AI-Based Astronaut Health Monitoring System

Project Proposal

Space missions expose astronauts to extreme conditions such as microgravity, radiation, and isolation, which can negatively impact their health. Monitoring astronaut health in real time is critical for mission success and safety.

This project proposes an Artificial Intelligence (AI) system that continuously monitors astronaut vital signs using wearable sensors. The system collects data such as heart rate, oxygen levels, and body temperature, and uses machine learning algorithms to detect abnormal patterns.

The AI system can alert astronauts and mission control if any health risks are detected early, allowing for quick intervention. This reduces the chances of serious medical emergencies during missions.

This solution is important for future long-duration missions, such as missions to Mars, where immediate medical assistance is not available. By using AI, NASA can improve astronaut safety and mission efficiency.